BatteryMonitor Windows

User Manual — v1.1

Overview

BatteryMonitor displays real-time battery data in two side-by-side charts and a two-row header. Controls are in the footer bar at the bottom.

The TC66 USB Power Meter (Optional)

BatteryMonitor optionally integrates with the MakerHawk TC66 USB Power Meter (~$26.99 on Amazon). The TC66 measures voltage, current, and power independently of the BMS, enabling accurate DC-DC charging efficiency calculations. It connects via USB serial (typically COM3–COM6 on Windows) and displays live V/i data on its own color LCD while simultaneously streaming data to BatteryMonitor.

> *Note: To reset the TC66's cumulative capacity (Qin) and energy (Ein) counters to zero, the physical button on the meter must be pressed. There is no software reset available.*

Footer Controls (left to right)

Control	Description
`Start / Stop`	`Begins or ends a recording session. While recording, the system is kept awake.`
`Clear`	`Clears all chart data and resets all accumulators without stopping the session.`
`Load CSV`	`Loads a previously saved CSV file for review. Live recording is suspended.`
`Int: N s`	`Sample interval in seconds (1–60). Lower values give finer resolution but larger files`
`Stop (red, TC66)`	`Disconnects the TC66 meter.`
`Span:`	`Time window shown in the charts (All, 1 h, 2 h, 4 h, 8 h). Does not affect recording.`
`Refr`	`Forces an immediate TC66 screen refresh.`
`TC66: COMx`	`Serial port selector for the TC66 USB meter.`
`Disc / Con`	`Connects or disconnects the TC66 meter on the selected port.`
■	`Flips the TC66 display orientation (useful when the meter is inverted).`
`Psys: N W`	`System baseline power (W) subtracted from TC66 input power when computing CE. Typicall`
`LR pts: N`	`Minimum number of SOC points used for the auto linear regression (TT5 prediction).`
`Run time`	`Elapsed time since Start was pressed.`
`N samples`	`Total number of samples recorded in the current session.`

Header — Row 0 (BMS data)

Row 0 is always visible. The status label on the left shows the current battery state:

State label	Meaning
CHRG	Charging
DCHRG	Discharging
IDLE	No significant current flow
FULL	Fully charged (trickle only)

During discharge (DCHRG) and charge (CHRG)

Field	Description
U=	Battery terminal voltage (V), 2 decimal places
i=	Discharge current (A), negative during discharge, 2 decimal places
R=	Internal resistance (Ω), integer
P=	Discharge power (W), negative; 2 decimals below 10 W, 1 decimal above
Qc=	Segment charge since last state transition (Ah)
Ec=	Segment energy since last state transition (Wh)
Qt=	Cumulative charge since Start (Ah)
Et=	Cumulative energy since Start (Wh)
SOC=	State of charge reported by BMS (%)
SoH=	State of health: ratio of current full-charge capacity to design capacity (%)

Header — Row 1 (TC66 data + computed)

Row 1 is visible when the TC66 is connected or when the TT5 countdown is active.

When TC66 is connected

Field	Description
U=	TC66 measured voltage (V)
i=	TC66 measured current (A)
R=	TC66 computed resistance (Ω), integer
P=	TC66 measured power (W)
Qin=	Cumulative charge delivered by TC66 since connection (Ah)
Ein=	Cumulative energy delivered by TC66 since connection (Wh)
CE=	Charging efficiency: Ec / (Ein - Psys x t) x 100 (%)
T.TC=	TC66 internal sensor temperature (°C) — not battery cell temperature

Always (when discharging)

Field	Description
TT5:	Estimated time remaining to 5% SOC, from auto linear regression on recent SOC data
Cyc=	Battery cycle count reported by BMS

Alerts

- Charge complete: when charging current drops to zero (CHRG→IDLE transition), a triple beep (880 Hz) sounds every 60 seconds for up to 10 minutes, signalling that a discharge cycle can be started. Dismissed by the red ✕ Beep button that appears on the right of the footer.
- Low SOC: a single beep (660 Hz) sounds at 10% SOC during discharge, then at each subsequent 1% step down to 1%.

Known BMS Limitations

- SOC step size varies by model: SOC is reported in discrete steps by the BMS; step size depends on the laptop model and SOC level, and is often smaller near full discharge than near full charge.
- BMS holdback (model-specific): on the ThinkPad T490, SOC was observed to hold artificially at 7% for 1–2 minutes before dropping sharply. This is a deliberate BMS algorithm to give the user time to save work, not a measurement artifact. It affects TT5 accuracy in this range. Other models may exhibit similar behavior at different SOC levels.
- Battery cell temperature is not accessible via any standard Windows API on the tested Lenovo models (Yoga 7 2-in-1, ThinkPad T490, ThinkPad X1 Carbon). T.TC= shows the TC66 meter's own sensor temperature, not the battery cell temperature.

The Two Charts

BatteryMonitor displays two dual-axis charts side by side, updated in real time at the selected sample interval. Both charts share the same elapsed-time X axis. Curve colors switch automatically between charge (green tones), discharge (red/orange tones), and idle (gray) states.

Chart 1 — Voltage, Current & Capacity

Left Y axis (U, V): battery terminal voltage and, when the TC66 is connected, TC66 input voltage.

Right Y axis (i, A & Qc, Ah): current and cumulative segment capacity.

Curve	Style	Description
i, A	Solid	BMS discharge/charge current (negative = discharge)
i.TC, A	Dashed	TC66 measured current
U, V	Solid	BMS battery terminal voltage
U.TC, V	Dashed	TC66 measured input voltage
Qc, Ah	Solid	Segment charge (resets on state transition)

Curve	Style	Description
Qt, Ah	Solid	Cumulative charge since Start

TC66 curves are only shown when the meter is connected.

Chart 2 — SOC, Power & Temperature

Left Y axis (P, W & Energy, Wh): power and cumulative energy curves.

Right Y axis (SOC, % & T, °C): state of charge and TC66 temperature.

Curve	Style	Description
SOC, %	Solid	State of charge reported by BMS
P, W	Solid	BMS power (negative = discharge)
P.TC, W	Dashed	TC66 input power
Ec, Wh	Dashed	Segment energy since last state transition
Et, Wh	Solid	Cumulative energy since Start
T.TC, °C	Dashed	TC66 internal temperature

Interactive Chart Features

Running Average (2-click)

Available on both charts. Click two points on any curve to display a running average line and value badge between them.

- Click 1: sets the start point (shown as a crosshair)
- Click 2: sets the end point; the average value is displayed
- Click to clear avg (shown in top-left of chart): removes the average

Linear Regression — TT5 Prediction (3-click, Chart 2 only)

The auto linear regression fits a line to recent SOC data and projects it forward to 5% SOC, giving the TT5 (Time to 5%) estimate shown in the header. A manual 3-click regression is also available:

- Click 1 & 2: define the SOC data range to fit
- Click 3: sets the target SOC (default 5%); the regression line and intercept time are displayed
- Click to clear LR (shown in top-right of chart): removes the regression line

The regression line color indicates fit quality:

- Green: $R^2 \geq 0.90$ — reliable prediction
- Orange: R^2 < 0.90 — poor fit, treat estimate with caution

Cursor Tooltip

Hovering over either chart displays a tooltip showing the time and value of the nearest data point across all visible curves.

Legend

The legend is moveable — click it to cycle through three positions (top-left, center, bottom-left). It lists all visible curves with their line style and color.

Time Window (Span)

The Span selector in the footer controls the time window displayed in both charts simultaneously. Options: All, 1 h, 2 h, 4 h, 8 h.

Installation & Requirements

Requirements

- Operating system: Windows 10 or Windows 11
- .NET Framework 4.0 or later (included in Windows 10/11 by default)
- Administrator privileges — required for WMI battery queries (health, cycle count)
- TC66 USB meter (optional) — for external V/i measurement and charging efficiency analysis

Installation

1. Download the latest release from the **Releases** page on GitHub.
2. Run the downloaded `BatteryMonitor_XX.Y.exe` directly — no installer or ZIP extraction required.
3. Ensure `app.manifest` is in the same folder as `BatteryMonitor.exe` (included in the release). To compile from source locally, `build.bat`, `app.manifest`, and the `.cs` source file must all be in the same folder.
4. Run `BatteryMonitor.exe` — a UAC prompt will appear; accept it to grant administrator access.
5. If Windows Smart App Control or Defender blocks the executable, you may need to right-click → Properties → Unblock, or disable Smart App Control in Windows Security settings.

First Run

- The app starts in idle state; click Start to begin recording.
- To use a TC66 meter, select the correct COM port and click Con before or after starting.

CSV File Format

CSV files are saved automatically to the application folder when CSV recording is active. Files are named BatteryLog_YYYYMMDD_HHMMSS.csv. A new row is written at each sample interval; the file is auto-saved every 10 seconds.

Column	Unit	Format	Description
Timestamp	—	yyyy-MM-dd HH:mm:ss	Wall-clock time of sample
Elapsed_s	s	F1	Elapsed time since Start
Voltage_V	V	F3	BMS battery terminal voltage

Column	Unit	Format	Description
Current_A	A	F3	BMS current (negative = discharge)
Power_W	W	F2	BMS power (negative = discharge)
SOC_Pct	%	F1	State of charge
State	—	text	Charging, Discharging, Idle, or Full
Qt_Ah	Ah	F6	Cumulative charge since Start
Et_Wh	Wh	F3	Cumulative energy since Start
Qc_Ah	Ah	F6	Segment charge since last state transition
Ec_Wh	Wh	F3	Segment energy since last state transition
TC66_V	V	F4	TC66 measured voltage (blank if no TC66)
TC66_A	A	F5	TC66 measured current (blank if no TC66)
TC66_W	W	F4	TC66 measured power (blank if no TC66)
TC66_Temp	°C	int	TC66 internal temperature (blank if no TC66)
TC66_Ah	Ah	F6	TC66 cumulative charge (blank if no TC66)
TC66_Wh	Wh	F3	TC66 cumulative energy (blank if no TC66)

TC66 columns are always present in the header but left blank for rows where no TC66 data was available.

About

Author	Tony Gozdz (tgozdz@gmail.com)
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Repository	https://github.com/ASG49/BatteryMonitor-Windows
License	MIT License
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